What representations guide learning to process singular they efficiently?

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Background: How do people learn to recognize singular 'they'?

Singular "they" has a new usage in English -- "personal they" -- specialized for people whose personal pronouns are they/them; often due to a nontraditional gender identity. "They" can also be plural.

Liz and Alex were at a restaurant. Liz handed the menu to Alex. Then <u>they</u> ordered some wine.

they = Alex or they = Liz and Alex?

How do people learn to recognize "they" as singular?

Mechanism 1: Top-down intentional decisions to use gender-fair language (Conrod, 2020)

Mechanism 2: Adaptation through exposure

Adaptation facilitates singular 'they' (Ye & Arnold, 2025)

<u>Singular exposure list (N=24):</u>

Alex was in the mailroom.

They picked up the mail. Then they borrowed scissors to open a package.

<u>Plural exposure list (N=24):</u>

Alex and Will were in the mailroom.

They picked up the mail. Then they borrowed scissors to open a package.

<u>Test sentence (N=20):</u>

Liz and Alex were at a restaurant.

Liz handed the menu to Alex.

Then they ordered some wine.

<u>20 Fillers with he/she/names</u>

<u>Task: click on the object mentioned (wine)</u>

Results:

- More singular responses on the singular exposure list (52%) vs. plural (28%)
- Responses are faster if they match the exposure

Research Question:

What representations are being primed?

Possibility 1: The singular or plural sense of "they" is primed.

For singular, maybe specifically the "personal they" usage

Possibility 2: The link *they* → Alex is primed.

Does exposure to *they* → *Alex* increase the chance of understanding "they" in the singular for other characters?

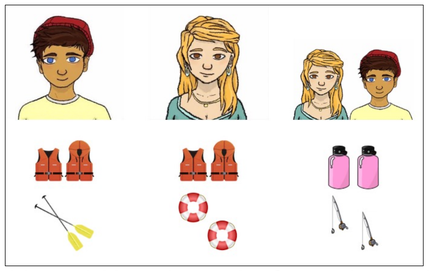
Methods

Story characters:

Liz (she) Will (he) Alex (they) Dana (they) Wren (they)

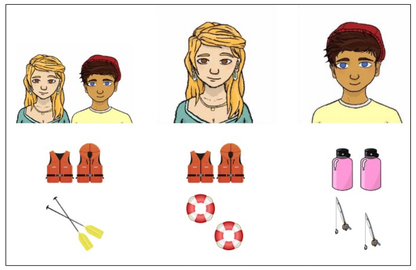
Exposure stories (between-subjects) all mention Alex as singular target

<u>Singular exposure list (N=24):</u>



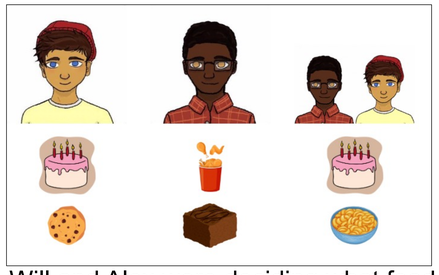
Liz and Alex planned a fun outing at the lake. Liz went kayaking with Alex. They lost their oars.

<u>Plural exposure list (N=24):</u>



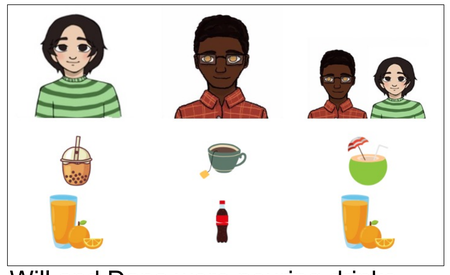
Test stories appear in two conditions (within-subjects):

<u>Primed target (Alex)</u>



Will and Alex were deciding what food to make for a party. Will gave an apron to Alex. They made a cake.

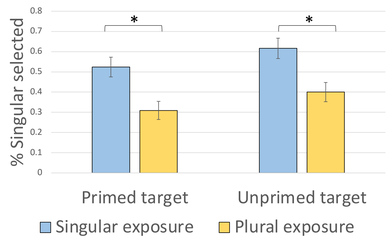
<u>Unprimed target (Dana/Wren)</u>



Will and Dana were pouring drinks. Will gave a cup to Dana. Then they had some orange juice.

Results

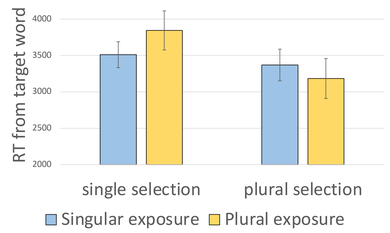
OBJECT SELECTIONS: More singular for singular vs. plural exposure



<u>Mixed effects logistic regression:</u>

- Main effect of exposure; pairwise tests show significant effects for both primed and unprimed targets
- No effects or interactions with target type

REACTION TIMES: Selections are faster when they match the exposure condition



<u>Mixed effects linear regression:</u>

- Interaction exposure x Selection
- Interaction exposure x Primed target
- Interaction exposure x Primed target

Unexpected interactions

May be due to item biases for primed and unprimed item sets (unprimed target stories were more singular-biased); longest RT for plural primed, singular selection, primed target (4423 ms)

	primed target	unprimed target
Singular exposure	3494	3407
Plural exposure	**3677**	3157

	primed target	unprimed target
Singular selection	**3959**	3385
Plural selection	3327	3175

Conclusions

- Singular *they* exposure facilitates singular interpretations for both familiar and new characters.
- Exposure activates general singular/plural representations (and may also activate character-specific information like "Alex uses *they*").

Participants:

77 subjects from the UNC subject pool; sample and analyses pre-registered

Thank you

Thank you to Kejia Zhang.

References

Conrod, K. (2020). Pronouns and Gender in Language. In K. Hall & R. Barrett (Eds.), The Oxford Handbook of Language and Sexuality (1st ed.). Oxford University Press.

Ye, Y. & Arnold, J. E. (2025). People adapt to Nonbinary Singular They through exposure. Poster, HSP 2025.

Poster